

NE8 Assignment 3: Serpent

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The serpent input file for all exercises can be found in `aqz22.Zhang_A_NE8_Assignment3.Serpent.Input.se`. This file is fully annotated with explanations of how the exercises were implemented in Serpent. I recommend using the Serpent 2 Syntax Highlighting extension for VSCode for readability.

All calculations for the input variables (such as fuel rod radii, densities of elements, etc.) were done in python. The code can be found in `aqz22.Zhang_A_NE8_Assignment3.inputcalculations.py`. The code for generating the plots can be found in `aqz22.Zhang_A_NE8_Assignment3.plots.py`. This program reads data from the `outputs.py` and `outputs8.py` files, which contain the Serpent output data. The code in these files is very basic so they will not be elaborated on any further.

There are several variables that were not given in the assignment. The following values were used: The density of UO_2 was set to 10.4 g/cm^3 , because that is the value used in Assignment 1 and Assignment 2. The density of H_2O is 0.66639 g/cm^3 , which was obtained by interpolating the *Density of Water and Steam* table in the CUED Thermofluids Data Book. Atomic densities were obtained from the IAEA Live Chart of Nuclides.

Unless otherwise stated, all Monte Carlo transport computations are performed with 100 inactive cycles, 1000 active cycles, and 10 000 source neutrons. These numbers exceed the recommended 20 inactive cycles, 500 active cycles, and 5000 source neutrons recommended in the Serpent Manual. The validity of these settings will be discussed in Exercise 8.

1 Using appropriate detectors, calculate radial distribution of U-238 capture reaction rate in the fuel pellet. Plot the U-238 capture rate versus radial distance from the fuel centre using 10 equal-volume radial regions. Explain the results.

For this exercise I will assume that the *capture reaction rate* refers to the macroscopic capture rate ($\text{MT} = -2$). The 10 equal volume radial regions were found using Equation 1.

$$r_i = R_{\max} \sqrt{i/10} \quad (1)$$

where $R_{\max}$ is the outer diameter of the fuel pellet, and r_i is the i -th equal-volume radius. The radial distribution of the macroscopic capture rate is shown in Figure 1.

Clearly, the capture rate increases with the radius. This can be explained by noting that the U-238 capture cross section is highest at the resonance peaks in the epithermal region. As a result, neutron capture tends to happen during thermalization, after the neutron has undergone one or a few scatter events. Neutrons are (mostly) slowed down by the H_2O moderator outside the pin. As these slower neutrons enter the pin from the outside, they will first be absorbed by the outer U-238. This leaves the inner U-238 with fewer neutrons at the absorption peaks. In summary, the capture rate is higher near the moderator, because inbound neutrons have a softer spectrum, while the absorption peaks are filtered out as these neutrons pass through the fuel rod.

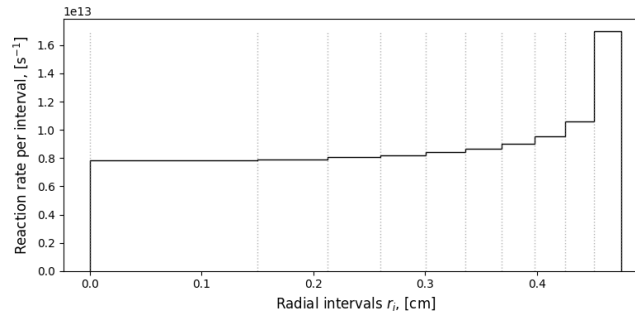


Figure 1: Radial distribution of the U-238 capture reaction rate in a 2D fuel pellet with radius 0.475 cm. Reaction rate is given per radius interval. Each radius interval represents a surface area of 0.070882 cm^2 .

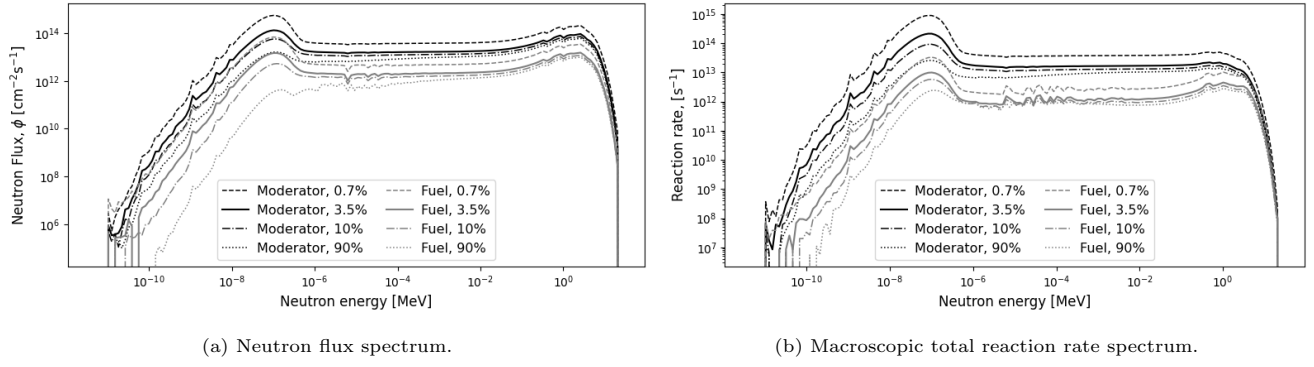


Figure 2: Graphs depicting energy spectra for the fuel and moderator regions with different levels of enrichment (shown in the legend). Vertical axes are the total amount inside one of 172 equal lethargy intervals between 10^{-6} eV and 100 MeV. Values are normalized to the lethargy width.

2 Plot the multi-group flux spectrum and macroscopic total reaction rate ($MT = -1$) in the fuel and moderator region. Obtain the flux spectrum in 172 equal lethargy groups (normalize the flux in each group to the group lethargy width $\Delta u_i = \ln(E_i/E_{i+1} + 1)$). Explain the results.

The neutron flux and macroscopic total reaction rate spectra are depicted by the solid lines in Figure 2. Looking at Figure 2a, 2 clear energy groups can be seen: the thermal group between 0.002 eV and 0.5 eV, as well as the fast group between 1 MeV and 10 MeV. The epithermal region between them is relatively flat. It represents the neutrons undergoing thermalization. The moderator reaction rate spectrum in Figure 2b has a much less pronounced fast neutron peak. This is an indication that H-1 has lower cross-sections than U-238 at high energy. A look at the cross sections of H-1 and U-238 in JANIS confirms this hypothesis.

Interestingly, the fuel spectrum clearly shows resonance peaks in the epithermal region. These peaks don't appear in the moderator spectrum. This result shows that H-1 does not have absorption peaks such as U-238, which is in agreement with JANIS data. Resonance peaks can also be found in the fuel curve of Figure 2a.

3 For the standard PWR fuel unit cell, calculate the U-238 capture ($MT = 102$) multi-group microscopic cross section in the fuel using the WIMS69 group structure. Do this for two separate fuel temperatures and relative contents of U-238 in the fuel. Describe the principles behind calculating multi-group cross sections and explain the results you obtain.

The U-238 microscopic cross sections for two different fuel temperatures and enrichments are depicted in Figure 3. The plot is large and uses log-linear axis to make the subtle differences between the curves more visible.

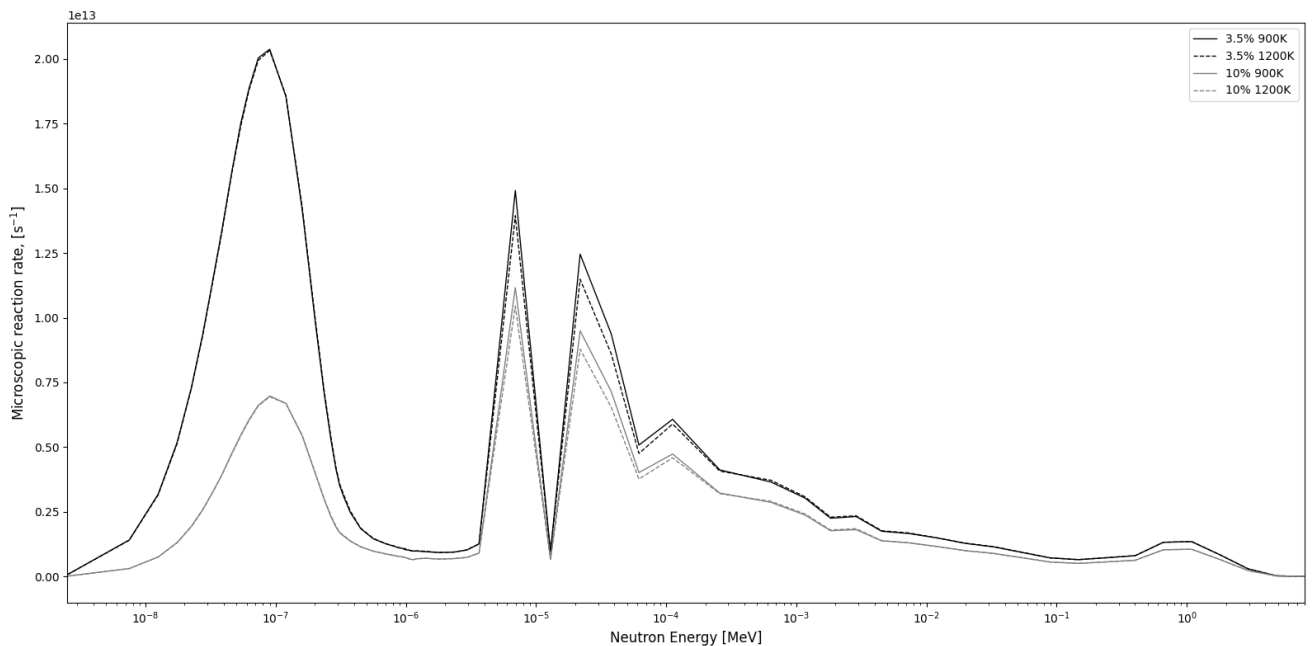


Figure 3: Graph showing the U-238 microscopic capture cross section in the fuel using the WIMS69 group structure, normalized to the lethargy width. Different curves represent different levels of enrichment and fuel temperatures, as shown in the legend.

The most obvious observation in Figure 3 is that a higher level of enrichment leads to fewer U-238 captures, especially in the thermal range. This result is caused by the reduced amount of U-238 atoms, as well as the increase in U-235 fission. The U-235 fission cross section is highest in the thermal range. Therefore, a higher enrichment level means that neutrons are much more likely to fission with the increased number of U-235 atoms. The reduction of U-238 atoms adds to this effect, because this reduces the chance of U-238 capture.

Close inspection of Figure 3 also reveals that an increase in temperature results in more high-energy captures, but fewer low-energy captures. Additionally, there is a net reduction in the total number of captures. These effects are all a result of the Doppler broadening of the U-238 absorption peaks. To start, the broadened peaks means that neutrons are more likely to be captured during thermalization, leading to an increase in high-energy captures. Next, since more neutrons are absorbed in the high end of the spectrum, there are fewer neutrons who make it to the lower energies. This is why the lower end of the spectrum has a lower capture rate. Lastly, the decrease in the resonance escape probability worsens the neutron economy, thus explaining the reduced reaction rate overall.

The principle for calculating multi-group cross sections relies on dividing the energy spectrum into *bins*. Bins are intervals of energy levels that are used for counting the number of reactions within that range. These bins are necessary because Monte-Carlo simulations are (in principle) continuous energy. During transport, every relevant interaction is tallied to corresponding energy bin. By averaging the tallies over many generations, the final tally is obtained. To increase the speed of convergence, tallying is only done when the fission source source has converged after a certain number of *inactive cycles*.

4 Calculate the DC, MTC and BW and explain the results.

The formula for calculating the reactivity coefficients was given:

$$\Delta\rho/\text{unit} = \frac{k_{\infty}(A_2) - k_{\infty}(A_1)}{k_{\infty}(A_1) \cdot k_{\infty}(A_2) \cdot [A_2 - A_1]} \cdot 10^5 \text{ [pcm/unit]} \quad (2)$$

where $k_{\infty}(A_1), k_{\infty}(A_2)$ are two values of k_{∞} obtained by a changing variable from A_1 to A_2 . The uncertainty propagation of Equation 2 is calculated using:

$$U\{\Delta\rho/\text{unit}\} = \sqrt{\left(\frac{10^5 \cdot U\{k_{\infty}(A_1)\}}{k_{\infty}^2(A_1) \cdot [A_2 - A_1]}\right)^2 + \left(\frac{10^5 \cdot U\{k_{\infty}(A_2)\}}{k_{\infty}^2(A_2) \cdot [A_2 - A_1]}\right)^2} \quad (3)$$

where $U\{x\}$ represents the uncertainty in variable x . For a derivation of Equation 3, refer to Appendix A. The perturbations used for finding $k_{\infty}(A_2)$ are: +5 °C for the MTC, +50 °C for the DC, and +50 ppm for BW. The results are shown in Equation 4.

$$\begin{aligned} DC &= -55 \pm 2 \text{ pcm/}^\circ\text{C} \\ MTC &= -1.3 \pm 0.2 \text{ pcm/}^\circ\text{C} \\ BW &= -31.3 \pm 0.2 \text{ pcm/ppm} \end{aligned} \quad (4)$$

As is typical, the reactivity is most sensitive to changes in the fuel temperature. This is why DC is the most important reactivity feedback in many reactors. The *MTC* has a lower absolute value but it is still negative. Considering the fact that coolant tends to heat up later than the fuel, this feedback is less significant, but important nonetheless. The *BW* represents the extent to which the reactor can be controlled by adding boron to the coolant. The negative value means that injecting more boron results in less reactivity. This is because boron is a strong neutron absorber.

Typical values for a PWR can be found here. Clearly, both *DC* and *BW* are significantly more negative than typical PWR values. The large *DC* may hamper reactor throttling.

5 Repeat the calculation of the multi-group flux spectrum in the fuel (like Q2) but for a variety of fuel enrichments: 0.7%, 3.5%, 10%, and 90%. Explain the results. Comment qualitatively on how simulation time changes and explain why, referring to the physics.

All of the neutron flux and macroscopic total reaction rate spectra are shown in Figure 2. In Figure 2a, lower enrichment levels have higher neutron flux. This may be surprising, as higher levels of enrichment have better neutron economies so they should have more flux. But this reasoning neglects the fact that power is normalized to 38 kW/kg_{HM}. Since higher enrichment leads to more fissions, fewer neutrons are needed to produce the same power. Therefore, a lower absolute number of neutrons is required when normalizing to power.

As for simulation time, lower enrichment levels took longer than higher enrichment levels. This can be explained by the way a Monte Carlo cycle works. Each generation starts with an empty fission bank. Every neutron is transported individually until it undergoes fission, where upon its offspring are placed in the fission bank. The cycle does not end until every neutron in the generation has undergone a fission interaction. Lower enrichment levels reduces the probability that an interaction is a fission. This means that neutrons needs to be transported for longer, thus increasing the duration of each cycle. The result is a longer simulation time for lower enrichments.

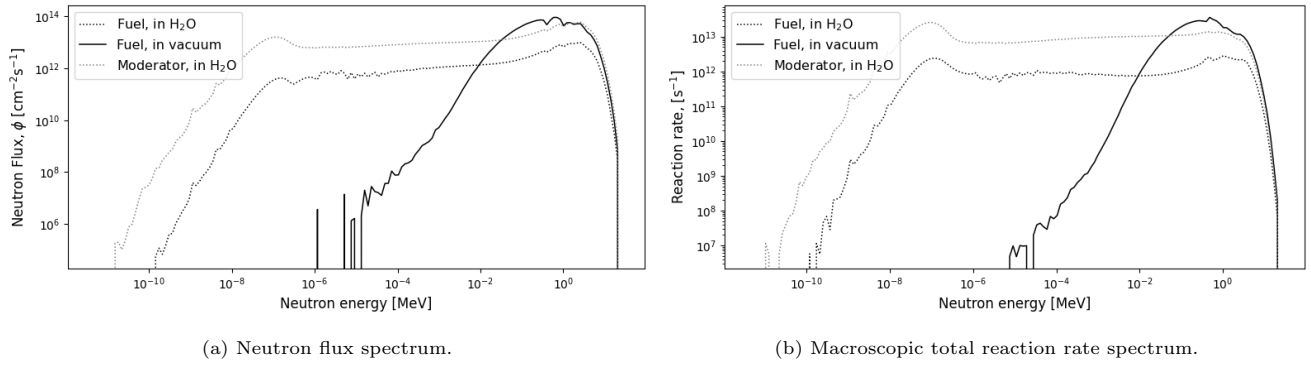


Figure 4: Graphs depicting energy spectra for the fuel surrounded by an H₂O moderator, and fuel surrounded by vacuum. Note that the vacuum fuel curves have numerical artifacts around 10⁻⁵ MeV. These have with a relative uncertainty of 100%.

6 Also calculate the flux spectrum in the case of 90% enrichment, but with vacuum instead of water. Comment on the differences between this spectrum and that of the moderated, 90% enriched design.

The neutron flux and total reaction rate spectra are shown in Figure 4. Obviously, the neutron spectrum is a lot harder when no moderator is present. Neutrons can only be slowed down by the oxygen, a terrible moderator. Neutrons with an energy below 10 eV aren't even present. This lack of moderation means that all the energy must be produced by fast fission. The lower fission cross section means that this is terrible for the neutron economy. Many more neutrons are needed to maintain the same power level. This is clearly visible by comparing the fuel curves in Figure 4a: The vacuum case requires orders of magnitude more neutrons.

7 Recalculate the coefficients for the moderated 90% enrichment case. Do they differ compared to those from question 5[sic]? Comment on the results.

The reactivity coefficients are calculated using the same method as explained in Exercise 4. The results are shown in Equation 5.

$$\begin{aligned} DC &= -1.6 \pm 0.7 \text{ pcm}/^\circ\text{C} \\ MTC &= -0.51 \pm 0.07 \text{ pcm}/^\circ\text{C} \\ BW &= -3.60 \pm 0.07 \text{ pcm/ppm} \end{aligned} \quad (5)$$

Comparing Equation 5 with Equation 4, shows that the reactivity coefficients are much less negative for the 90% enrichment case. This is a result of two factors: the decrease in U-238 absorption and the increase in U-235 fission. These two changes mean that the relative effect of U-238 is much less important. This explains why the *DC* is so much lower: U-238 absorption plays a much smaller role in high-enrichment fuels.

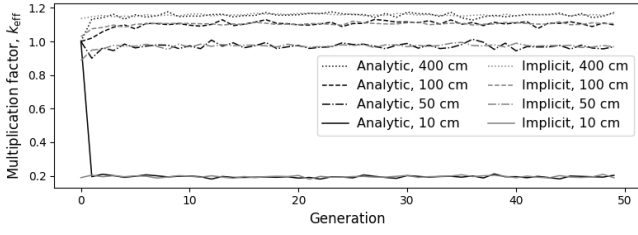
The increase in U-235 fission also means that much more neutrons fission while they are still fast. Thus, the spectrum hardens as a whole. This is another reason why the *DC* is lower: there are fewer epithermal neutrons. Additionally, the increase in fast fissions means that moderation is much less important, which is why the *MTC* is lower. Lastly, the *BW* is lower because the boron capture cross section is high in the thermal range. The hardened spectrum means that the effect of boron capture is greatly reduced.

8 MONEY QUESTION (2 parts)

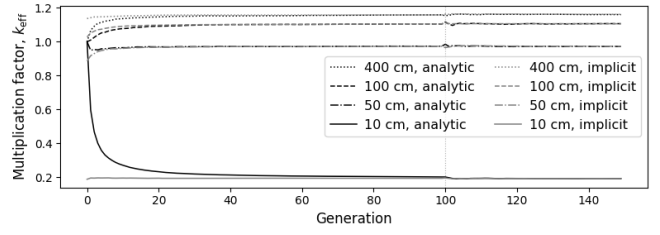
8.1 Convert the 2D pin to a 3D pin with reflective radial boundary conditions and vacuum axial boundary conditions. Use the 'set his' and 'set entr' options in Serpent to measure k_{eff} and Shannon entropy over cycles. Describe how the convergence rate varies with different axial dimensions between 10cm and 4m, and comment on the two metrics.

The convergence of k_{eff} is shown in Figure 5. Two different methods are depicted: The analytic method is derived from the amount by which the number of simulated neutrons were multiplied. The implicit method is derived from the expected number of neutrons produced given the cross sections of the interactions. Looking at Figure 5, the implicit method converges much more quickly than the analytic method. This is because the implicit method takes more interactions into account, not just the ones that produce neutrons for the next Monte Carlo cycle.

A surprising result can be found in Figure 5a, as there does not appear to be any correlation between the axial length and the speed of convergence. This is because all simulations converge extremely quickly: after just one round. This fact means that in Figure 5b, for the analytic method, the values closest to the initial guess of 1 converge the fastest. In contrast, the implicit method does not rely on any initial guess.

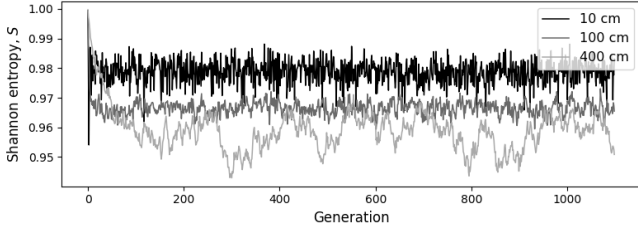


(a) Graph depicting the specific k_{eff} values obtained during each cycle.

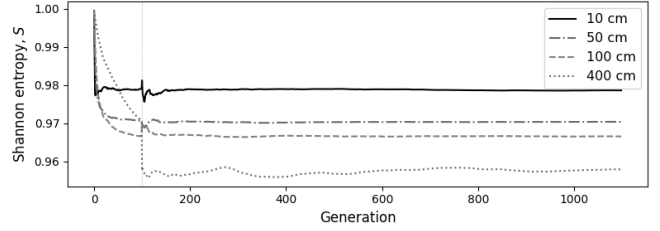


(b) Graph depicting cumulative average k_{eff} values. Vertical line indicates the first active cycle after 100 generations.

Figure 5: Graphs depicting the measurements of k_{eff} over different Monte Carlo cycles of 3D fuel pin. The curves represent different axial lengths and different ways of obtaining k_{eff} , as shown in the legend.



(a) Graph depicting the specific entropy values obtained during each cycle.



(b) Graph depicting cumulative average entropy values. Vertical line indicates the first active cycle after 100 generations.

Figure 6: Graphs depicting the measurements of the Shannon entropy over different Monte Carlo cycles of 3D fuel pin. The entropy mesh consists of 1 radial and 25 axial bins. The curves represent different axial lengths, as shown in the legend.

If k_{eff} is the only value of interest, it can be concluded that the number of inactive cycles can be greatly reduced. A value close to 1 suffices.

In the case of Monte Carlo neutron transport, Shannon entropy can be interpreted as the amount of information each generation provides about the system. It measures this by looking at how predictable the fission source distribution is. Figure 6 shows the convergence of the Shannon entropy. It is important to note that all these simulations were done with the same number of neutrons. This time, there are clear differences in the speed of convergence. The longer axial lengths converge more slowly than the shorter lengths. This makes sense because Shannon entropy is derived from the fission source. For longer axial lengths, neutrons need more generations to diffuse over the longer distance before the source converges. Another conclusion is that, the source distribution converges more slowly than k_{eff} . It can also be observed that longer axial lengths have a lower Shannon entropy. This means that less information is gained about the source distribution per generation. This might be because the number of neutrons was the same for all axial lengths, while longer lengths should require more neutrons to cover more volume. It appears that 100 is about the right number of inactive cycles for the 400 cm fission source to converge.

A more subtle observation comes from Figure 6a. It appears that the 400 cm case never quite “settles”, but has persistent multi-generational fluctuations around some average value. This is likely caused by *neutron clustering*, which is a tendency for neutrons to “clump” together over multiple generations. This happens due to an imbalance between the birth of neutrons, which happens near a parent neutron, and the death of neutrons, which is spatially uniform. These neutron clusters are deviations from the fundamental mode distribution: they would not appear if the number of neutrons was sufficiently high. As a consequence, these neutron clusters decrease the amount of information that can be gained in each generation. This is clearly shown in Figure 7, which depicts the entropy when the number of neutrons is increased. Note that the inter-generational fluctuations diminish when there are more neutrons. However, 100 000 neutrons was very computationally expensive. For more information on neutron clustering and entropy refer to: Nowak et al. *Monte Carlo power iteration: Entropy and spatial correlations* (2016).

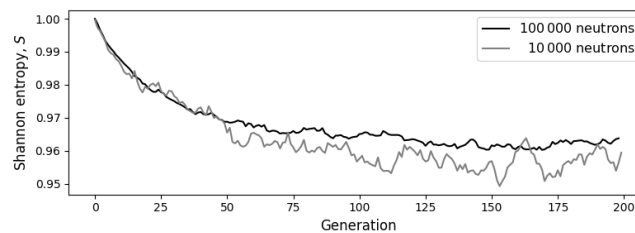
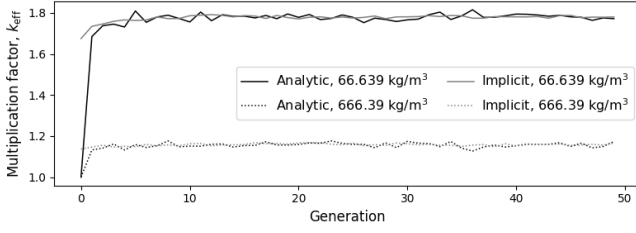
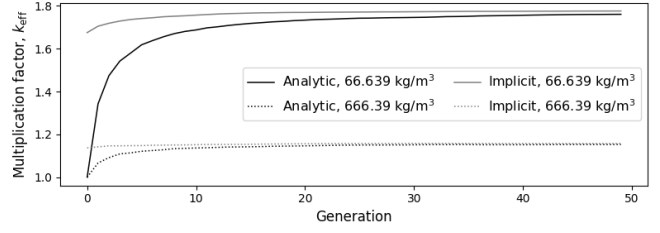


Figure 7: Graph comparing the cycle-to-cycle entropy of a 400 cm fuel rod using 10 000 or 100 000 neutrons in the Monte Carlo simulation.

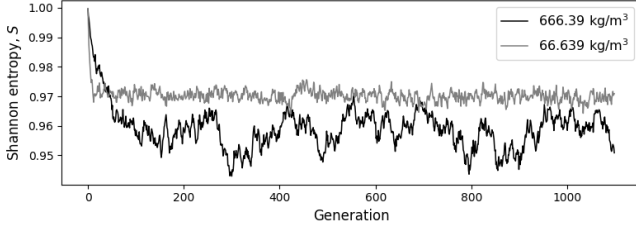


(a) Graph depicting the k_{eff} values obtained during each cycle.

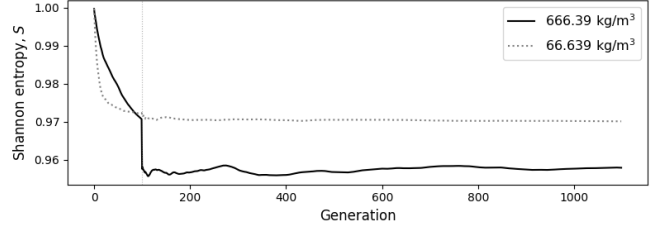


(b) Graph depicting cumulative average k_{eff} values.

Figure 8: Graphs depicting the measurements of k_{eff} over different Monte Carlo cycles of 3D fuel pin. The curves represent different water densities and different ways of obtaining k_{eff} , as shown in the legend.



(a) Graph depicting the specific entropy values obtained during each cycle.



(b) Graph depicting cumulative average entropy values. Vertical line indicates the first active cycle after 100 generations.

Figure 9: Graphs depicting the measurements of the Shannon entropy over different Monte Carlo cycles of 3D fuel pin. The entropy mesh consists of 1 radial and 25 axial bins. The curves represent different water densities, as shown in the legend.

8.2 Now decrease the water density by 10[sic] in the 4m case. How does the convergence rate compare to the 4m full density case? Explain why.

Note that the exercise contains a typo: the water density was decreased by *a factor of 10*. The convergence of k_{eff} is compared in Figure 8. There is no clear difference in the speed of k_{eff} convergence, because the value converges so quickly that any difference is not visible.

The entropy convergence is shown in Figure 9. It can be concluded that the source distribution does converge much more quickly when the water density is low. The amount of neutron clustering is also greatly reduced. As explained before, the higher entropy indicates that fewer neutrons are needed to obtain information about the system. There are several reasons for this. Firstly, a lower water density means a hardened spectrum. This means that a smaller portion of the spectrum needs to be “explored” by the neutrons, thus fewer neutrons are needed. Secondly, the reduction in water density also reduces scattering, so the neutrons are able to travel further in each generation. This allows neutrons to explore and gain information on a larger volume. Thirdly, the increased neutron diffusion also means that neutron clusters are dispersed more easily. The lack of neutron clusters means that neutrons are more evenly distributed through the whole system, and thus gain information more efficiently. There may be more reasons for the higher entropy, but they should all rely on this same principle of neutrons gaining more information.

A Uncertainty of Reactivity Coefficient

The reactivity coefficient is calculated from 2 dependent variables: $k_{\infty}(A_1)$ and $k_{\infty}(A_2)$. Note that A_1 and A_2 are independent variables used as inputs, so they have no uncertainty. Therefore the uncertainty propagation relation is:

$$U\{\Delta\rho/\text{unit}\} = \sqrt{\left(\frac{\partial\Delta\rho/\text{unit}}{\partial k_{\infty}(A_1)}U\{k_{\infty}(A_1)\}\right)^2 + \left(\frac{\partial\Delta\rho/\text{unit}}{\partial k_{\infty}(A_2)}U\{k_{\infty}(A_2)\}\right)^2} \quad (6)$$

Taking the derivative of Equation 2 with respect to $k_{\infty}(A_1)$ results in:

$$\frac{\partial\Delta\rho/\text{unit}}{\partial k_{\infty}(A_1)} = \frac{-10^5}{k_{\infty}^2(A_1) \cdot [A_2 - A_1]} \quad (7)$$

Mutatis mutandis for $k_{\infty}(A_2)$.

$$\frac{\partial\Delta\rho/\text{unit}}{\partial k_{\infty}(A_2)} = \frac{-10^5}{k_{\infty}^2(A_2) \cdot [A_2 - A_1]} \quad (8)$$

Substituting both partial derivatives into Equation 6 results in:

$$U\{\Delta\rho/\text{unit}\} = \sqrt{\left(\frac{10^5 \cdot U\{k_{\infty}(A_1)\}}{k_{\infty}^2(A_1) \cdot [A_2 - A_1]}\right)^2 + \left(\frac{10^5 \cdot U\{k_{\infty}(A_2)\}}{k_{\infty}^2(A_2) \cdot [A_2 - A_1]}\right)^2} \quad \blacksquare \quad (9)$$